Machine Protection Door

ASSA ABLOY RP300 / RP300 Wide / RP300 USD

ASSA ABLOY

ASSA ABLOY Entrance Systems

The global leader in door opening solutions



HIGH SPEED

Rapid opening and closing ensures short cycle time.



HIGH RELIABILITY

High reliability and low service even after years of high numbers of cycles.

HIGH SAFETY

Safety contact edge or touchless safety edge. Safety limit switches according to PLe and SIL3.

ASSA ABLOY as word and logo are trademarks owned by the ASSA ABLOY Group.

Machine Protection Door ASSA ABLOY RP300 / RP300 Wide / RP300 USD

Advantages of the door

- Rapid opening and closing ensures short cycle times up to 5 cycles per minute
- High reliability even after years of high numbers of cycles
- Safety limit switches according to EN ISO 13849-1 and EN 62061
- Doors conform to EN ISO 12100 and EN ISO 14119
- Versions conforming to various factory standards are possible
- Type tested machine protection door with fabric door blade

Design

Side Frames
The side frames are made of aluminum profiles. The front cover side frame can easily be removed and allows easy access for inspection and maintenance.

Windows
For process monitoring when the door is closed, various window variants are available. For all other variants of curtains, optional horizontal or vertical windows are integrated in the door curtain. Also anti-UV windows conform to EN 1598.

Drive Unit
Gear motor with safety gear box including catching device, mounted on the right or left side.

Control System
Depending on the application there are 2 control systems which offer a broad spectrum of speeds, power supplies and options for connecting control and safety devices.

Standard: ACS 50
Option: frequency converter control MCC

Upon request the door can also be delivered without a control system (not Upside Down). Extension box: Space saving extension box for MCC allows mounting of interface card or safety monitoring module for safety limit switches.

Safety Limit Switch integrated in Side Frame
For the indication "door safely closed", safety limit switches are integrated in the side frame.
Optional evaluation by safety systems of the machine (version without safety monitoring module).
Due to the use of safety limit switches, machine protection doors cannot be equipped with a knock-out mechanism.

Model ASSA ABLOY RP300 USD
The new ASSA ABLOY RP300 USD (Upside Down) machine protection door is designed so that you can integrate it into your production area where bigger components and machinery must be arranged vertically.

CE Guidelines
This door is designed according to the regulations of the Workplace Directive of the UVV as well as the harmonized CE Guidelines including the EN 13241-1 and 2006/42/EG.

Safety Features
The bottom profile is monitored by an electrical safety contact edge. As an option, the door can be equipped with a pre-running photocell in combination with the control system MCC (Standard on Upside Down). A door line photocell also prevents the door from closing when objects are interfering.

Integrated Cable Chain
Integrated cable chain guides cables for pre-running photocell or electrical safety contact edge inside the side frame.

Indication of Risk
The security of the door is designed for normal use for vehicles in industrial environments corresponding to the harmonized CE-Guidelines. In addition, special environmental conditions may impact the type of door necessary. For clarification and details, please contact your local sales engineer.

Door Curtain
From partially transparent PVC curtains with colored reinforcement stripes with a high degree of transparency to highly rigid RollTex® and NomaTex®. All door curtains are silicon-free.
Side-mounted wind tabs reduce inward bending of the door blade. RollTex® according to TRAV (technical rules for use of anti-fall guard vitrification).

RP300 **RP300 wide** (Color code according to RAL color card)	PVC with colored fabric reinforcement strips	PVC with colored fabric reinforcement strips silicone-free	RollTex® Plus* TRAV	RollTex® Original* TRAV	NomaTex®
Windows	–	–	•	•	•
Vision panels	–	–	–	–	•
Blue	• 5002	• 5002	✓ 5002	• 5002	• H 5005
Red		• 3002			• NA 3000
Orange		• 2011	• 2009	• 2009	• H 2009
Grey	• 7035	• 7037	• 7001	• 7001	• 7038
Yellow		• 1021			• NA 1023
Green	• 6018				
Beige	• 1014				
White	• 9016	• 9010			
Black		• 9005			

H = Smooth side on the front NA = not antistatic

RP300 USD only „RollTex® Original"

Technical Data ASSA ABLOY RP300 / RP300 Wide / RP300 USD

TECHNICAL DATA	ASSA ABLOY RP300	ASSA ABLOY RP300 Wide	ASSA ABLOY RP300 USD
Application	Interior door / Machine Protection Door	Interior door / Machine Protection Door	Interior door / Machine Protection Door
Door dimensions (mm) (DW min./max.) (DH min./max.)	 750 / 4000 mm 750 / 3500 mm	 750 / 6000 mm 750 / 5000 mm	 1000 / 5500 mm 1000 / 3500 mm
max. area	14 m²	30 m²	14 m²
Opening direction	vertical	vertical	vertical
Wind resistance (EN 12424)	class 1 (max. 11 m²)	class 1 up to DW ≤ 5600 mm	class 1
Noise emission	< 70 dB A	< 70 dB A	< 70 dB A
Temperature range max. (Extended on request) [3]	+10°C – +45°C (0 up to +60°C)"	+10°C – +45°C (0 up to +60°C)"	+10°C – +45°C (0 up to +60°C)"
Surface material			
Side frame	✓ Anodized aluminum	✓ Anodized aluminum	✓ Anodized aluminum
Top roll cover	• Anodized aluminum	• Anodized aluminum	✓ Anodized aluminum
Bottom profile	✓ Anodized aluminum	✓ Anodized aluminum	✓ Anodized aluminum
Motor cover	• Powder coated steel, RAL 9006	• Powder coated steel, RAL 9006	• Powder coated steel, RAL 9006
Extension side frame	–	–	250 – 500 mm (depending on doortype)
Safety features			
Electrical safety contact edge	✓	✓	–
Door line photocell	✓	✓	–
Pre-running photocell (only in combination with MCC, Door line photocell as option)	•	•	✓
Drop down protection in drive unit	✓	✓	–
Pre-running photocell [2]	–	–	✓
Emergency egress	Crank handle	Crank handle	Crank handle
Counterbalance	–	–	Counterbalance
Equipment			
LED traffic light	•	•	•
Traffic light / Flashing alarm light	•	•	•
Suitable machine protection	✓	✓	✓
Magnetic switch in side frame	✓ 1 piece / • 2 piece	✓ 1 piece / • 2 piece	✓ 1 piece / • 2 piece
Window in sideframe, view on magnetic switch	•	•	•
Drive unit			
Motor power	0,75 kW	0,75 kW	0,75 kW
Chain drive	•	•	–
Gear heating	•	•	•
Control system	✓ ACS 50 / • MCC	✓ ACS 50 / • MCC	✓ MCC
Control system			
MCC Open/Close max. (m/s) [2]	• 2,4 / 1,3 2,4 / 1,75 (pre-running photocell)	• 1,7 / 1,0 1,7 / 1,1 (pre-running photocell)	✓ 2,2 / 1,5
ACS 50 Open/Close max. (m/s) [2]	✓ 1,0 / 1,0	✓ 0,8 / 0,8	–
Cycles per minute max. * exact number of cycles depending on application	ACS 50 = 1 MCC = 3 up to 5*	ACS 50 = 1 MCC = 1 up to 1,5*	MCC = 5*
Separate display / control	only at MCC	only at MCC	only at MCC
Fuse protection by site	10 A (C-characteristic) (at 220/230V 16 A)	10 A (C-characteristic) (at 220/230V 16 A)	10 A (C-characteristic) (at 220/230V 16 A)
Control voltage	24 V DC	24 V DC	24 V DC
Protection	IP 55	IP 55	IP 55
Potential-free contacts	•	•	•
UL-Version	• ACS50 • MCC	• ACS50 • MCC	• MCC
Network connection			
ACS 50	Standard: 3L/(N)PE/220/230/380/400/415V; 50Hz with transformer: 3L/(N)PE/440/480/500V; 50Hz		–
	UL-Version: 3L/(N)PE/208/220/230/460/480V; 60Hz		–
MCC	Standard: 3L/(N)PE/380/400/415/440/460/480V;50/60Hz with transformer: 3L/(N)PE/220/230/500V;50Hz		
	UL-Version: 3L/(N)PE/208/220/230/460/480V; 60Hz		

• Option ✓ Standard *) Specific number of cycles on request

1) For doors of height < 2300 mm the use of top roll cover is required according to standard EN 13241-1.
2) Deviations possible depending on other properties
3) Depending on cycles/minute, door size and installation location



Saftey limit switch
Optional: For the indication "door safely closed", safety limit switches are integrated in the side frame.



Floor mounting brackets
The door can be mounted free-standing on floor with optional mounting brackets. Levelling screws allow adjustment by uneven floor.



Motor position
Motor can be mounted right- or lefthand side. Motor position can be chosen in step of 45°.
A chain drive is available as option in case of space limitations.



Example of use: door in assembly line

The demands in automated manufacturing processes are high and the smallest interruption or delay can break down the flow of production or endanger employees. Machine protection doors help manufacturers avoid both scenarios. The doors are integrated into the processes and open and close according the production run. They enable the rapid change from complete isolation of the production step to free access to work piece and machine within seconds. It allows short cycle times and can be installed in machines, on production lines and railing systems within close proximity to the machine. Plus, they protect people in highly automated areas, making processes much safer.

Advantages of the ASSA ABLOY RP300/RP300 Wide/RP300 USD

- Service friendly: Two-pieced aluminum side frames offer easy access for inspection and maintenance
- Pre-running photocell: in combination with control system ASSA ABLOY MCC a pre-running photocell is available as option. It provides more comfort and higher closing speed
- Integrated cable chain: The cables of the electrical safety contact edge and pre-running photocell are guided in cable chains inside the side frame
- Pluggable cables in the cable chain
- Drive unit: A chain drive is available as option in case of space limitations

Connection with the machine

The in- and outputs of the door control unit MCC (potential 24 V DC) can be connected directly, to the machine. The in- and outputs of the door control unit MCC and ACS 50 can be connected potentialfree with the machine. As standard the following signals are available:

- 2 potential-free safety contacts, when the door is closed (safety limit switch)
- 1 output or potential-free signal, when the door is open
- 1 output or potential-free signal when the door is closed
- 1 output or potential-free signal for faults

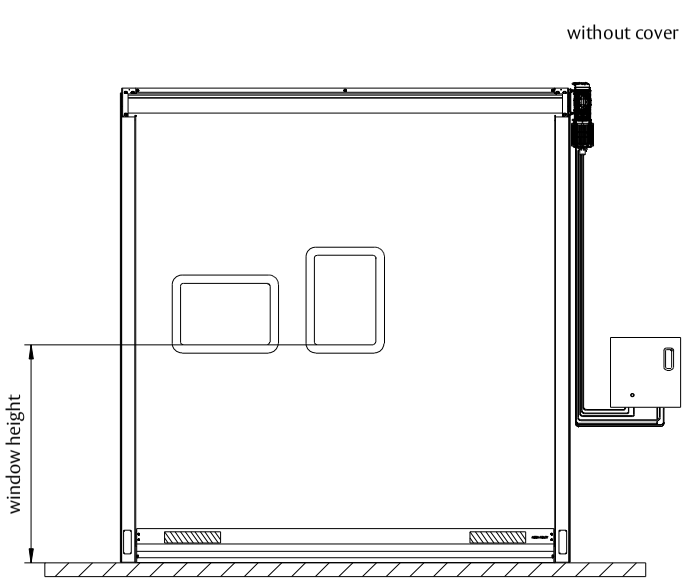


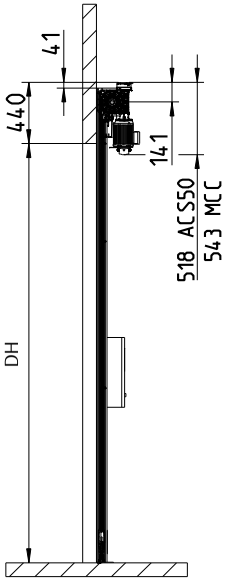
The ASSA ABLOY RP300 USD closes from bottom to top

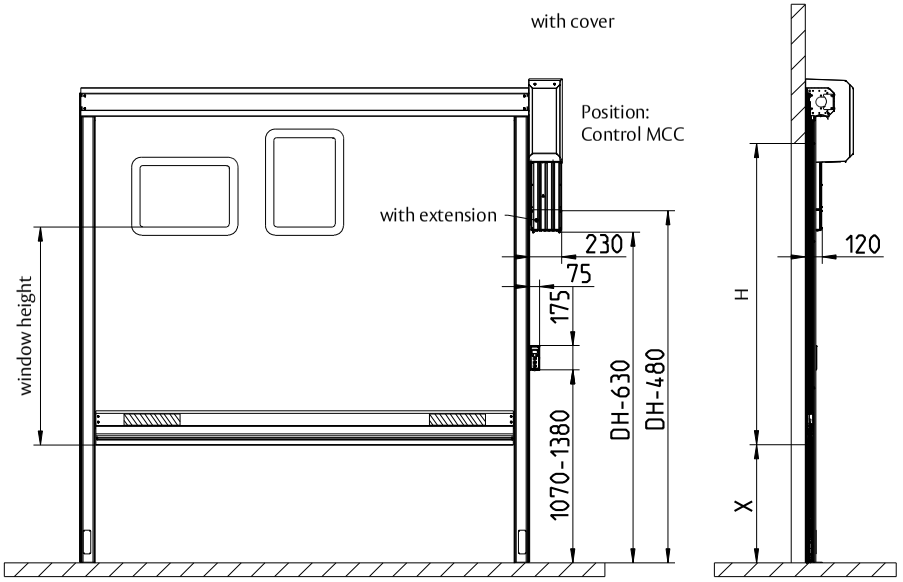


Example of use: place a workpiece into the process

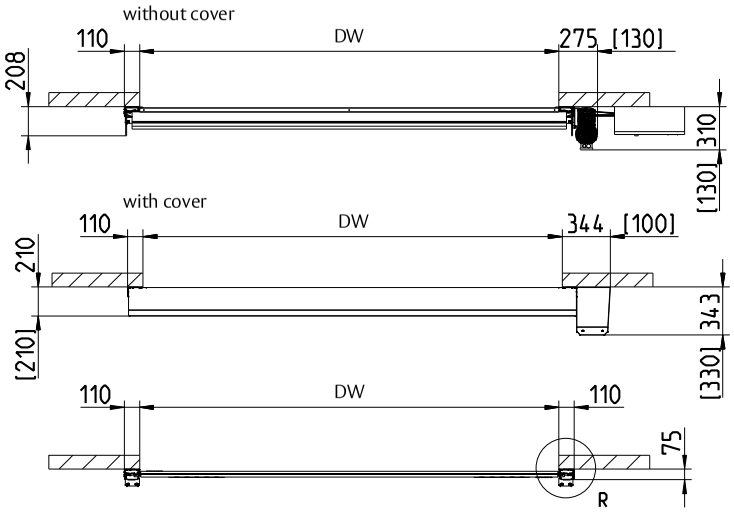
General drawing ASSA ABLOY RP300

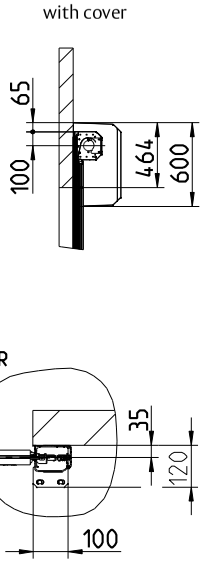


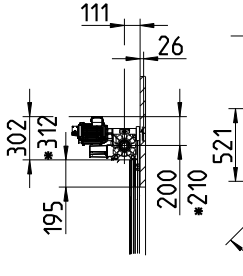




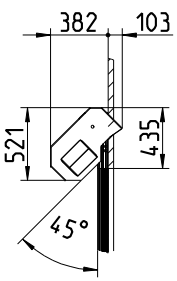
„Option X-Level“
Door does not close to floor, but down to X-Level



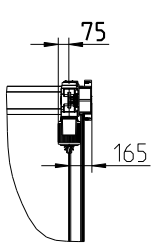




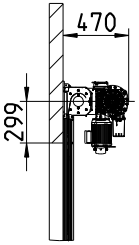
every 45°

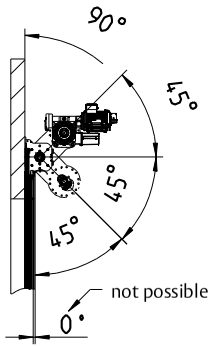


every 45°

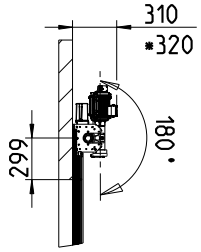


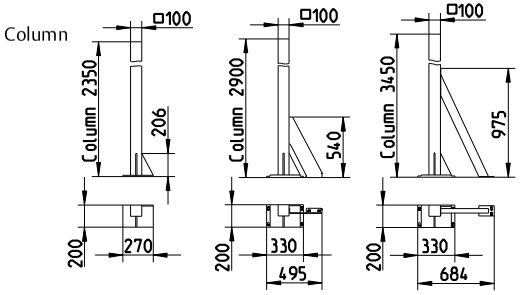
Chain drive





not possible





drive unit- and chain drive position not drawn at request

Tolerance ± 3 mm (for all dimensions)
Dimensions in [ ] = clearance for installation

* UL-drive unit

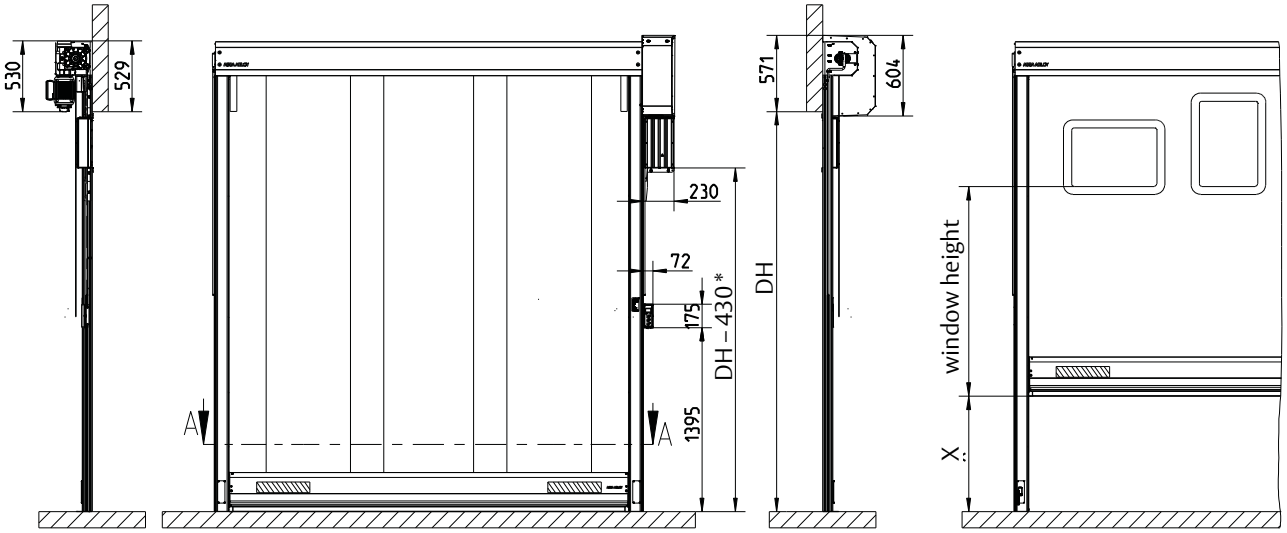
9300R0001/e

General drawing ASSA ABLOY RP300 Wide

without cover

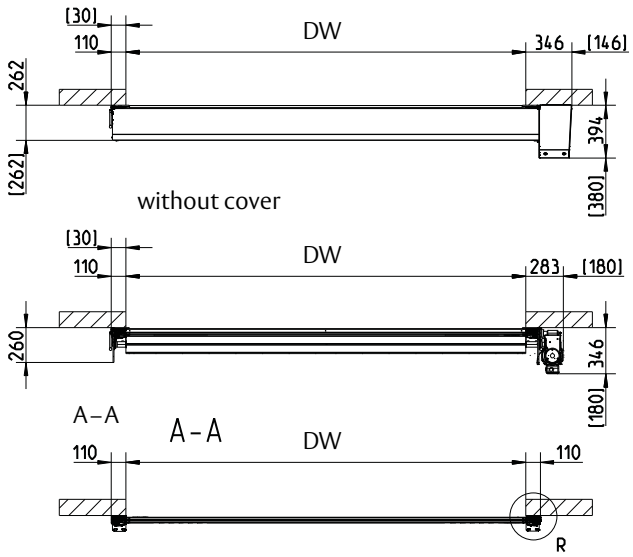
with MCC at side frame

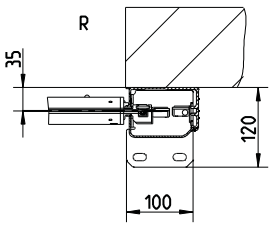
with window



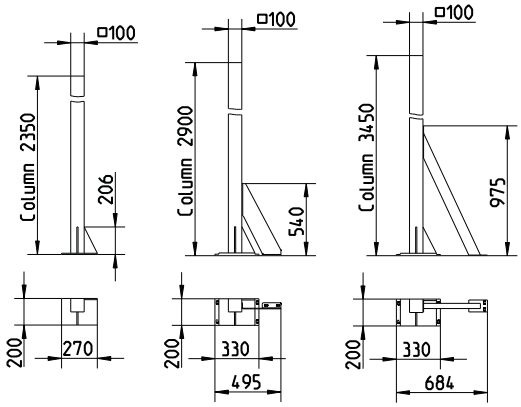
* DH – 590 with extension

X = X-Level





Column

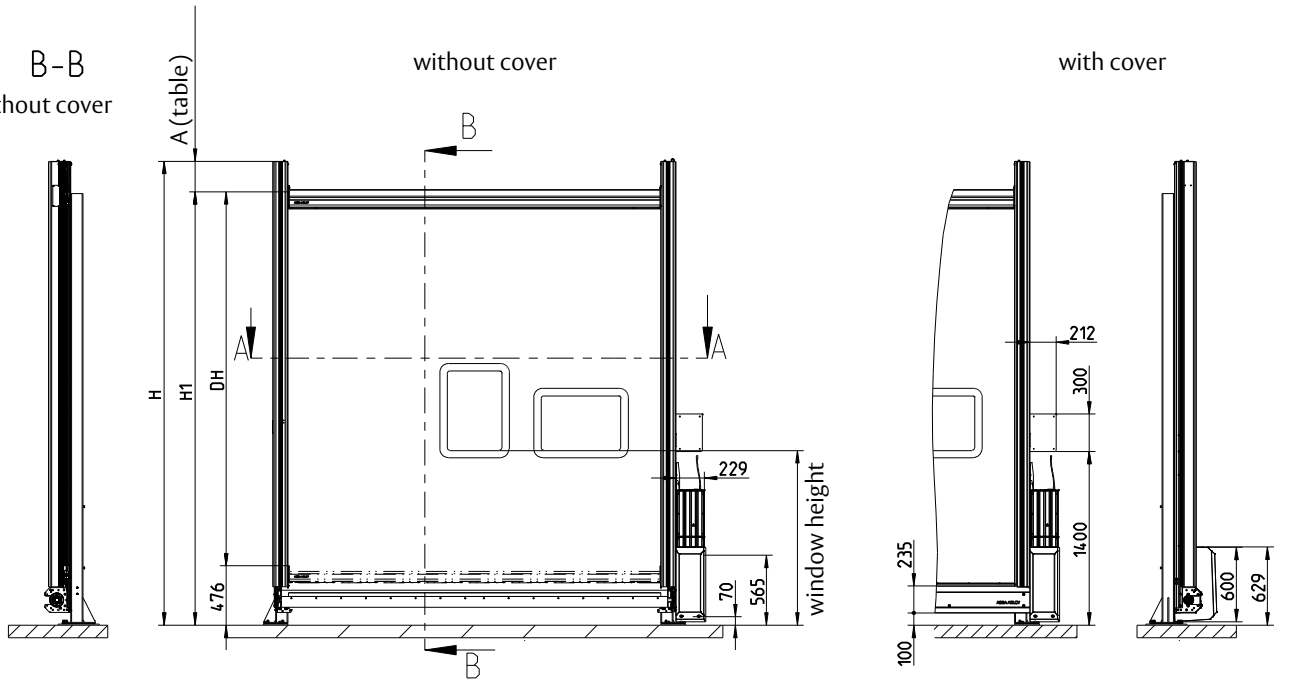


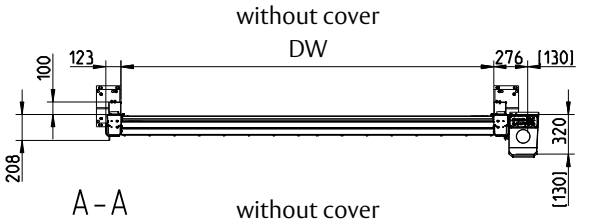
Available sizes		
	DW (B)	DH (H)
min. (mm)	750	750
max. (mm)	6000	5000
DW > 4000 or DH > 3500		

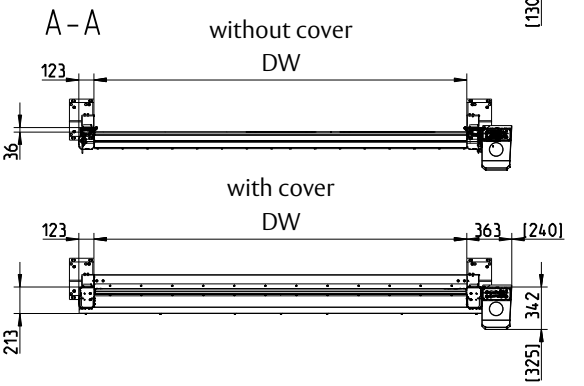
Control Unit			
		ACS 50	MCC
Dimensions (W × H × D)	mm	300 × 400 × 210	220 × 380 × 100

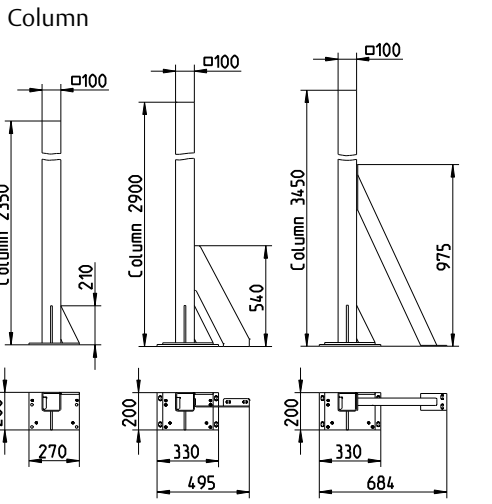
[ ] = Space needed for installation

General drawing ASSA ABLOY RP300 USD

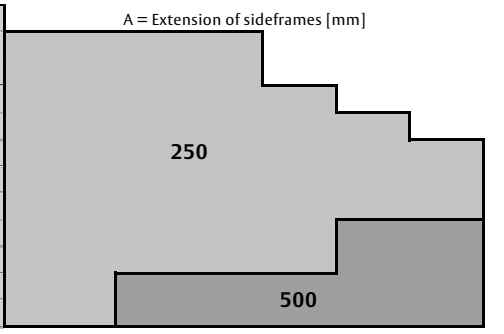








Height [m]
3,500
3,250
3,000
2,750
2,500
2,000
1,500
1,375
1,250
1,125
1,000



Available sizes		
	DW	DH
min. (mm)	1000	1000
max. (mm)	5500	3500

[ ] = Space needed for installation







ASSA ABLOY Entrance Systems is a leading supplier of entrance automation solutions for the efficient flow of goods and people. Building on the long-term success of the Besam, Crawford, Albany and Megadoor brands, we offer our solutions under the ASSA ABLOY brand. Our products and services are dedicated to satisfying end-user needs for safe, secure, convenient and sustainable operations.
ASSA ABLOY Entrance Systems is a division of ASSA ABLOY.

assaabloyentrance.com



- Production Location
- Sales Organisation
- Reseller

For subsidiary or representatives in your region visit:
assaabloyentrance.com

ASSA ABLOY Entrance Systems

assaabloyentrance.com

Follow us:

Please enter *ASSA ABLOY Entrance* in the channel's search field.

© ASSA ABLOY Entrance Systems AB BR-MAN/ASSA ABLOY RP300 Wide/de_EN-1.1/1802/64 100 728 Technical data subject to change without notice